University of Antwerp
Faculty of Science

Computer Networks (2025-2026)
Part 0: Course Introduction

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My academic journey

Jeroen Famaey

https://jeroen.famaey.eu

2003 – 2007 **Master Informatics**
Ghent University, Belgium

2007 – 2012 **PhD in Computer Science Engineering**
Ghent University, Belgium

2012 – 2014 **Postdoctoral Researcher (video streaming and content delivery)**
Ghent University

2014 – 2015 **Postdoctoral Researcher (wireless networks)**
University of Antwerp

2016 – now **Research Professor (wireless networks and embedded computing)**
IDLab, University of Antwerp, Belgium

- *Computernetwerken (BA2)*
- *Computer- en netwerkbeveiliging (BA3)*
- *Advanced Networking Lab (MA)*

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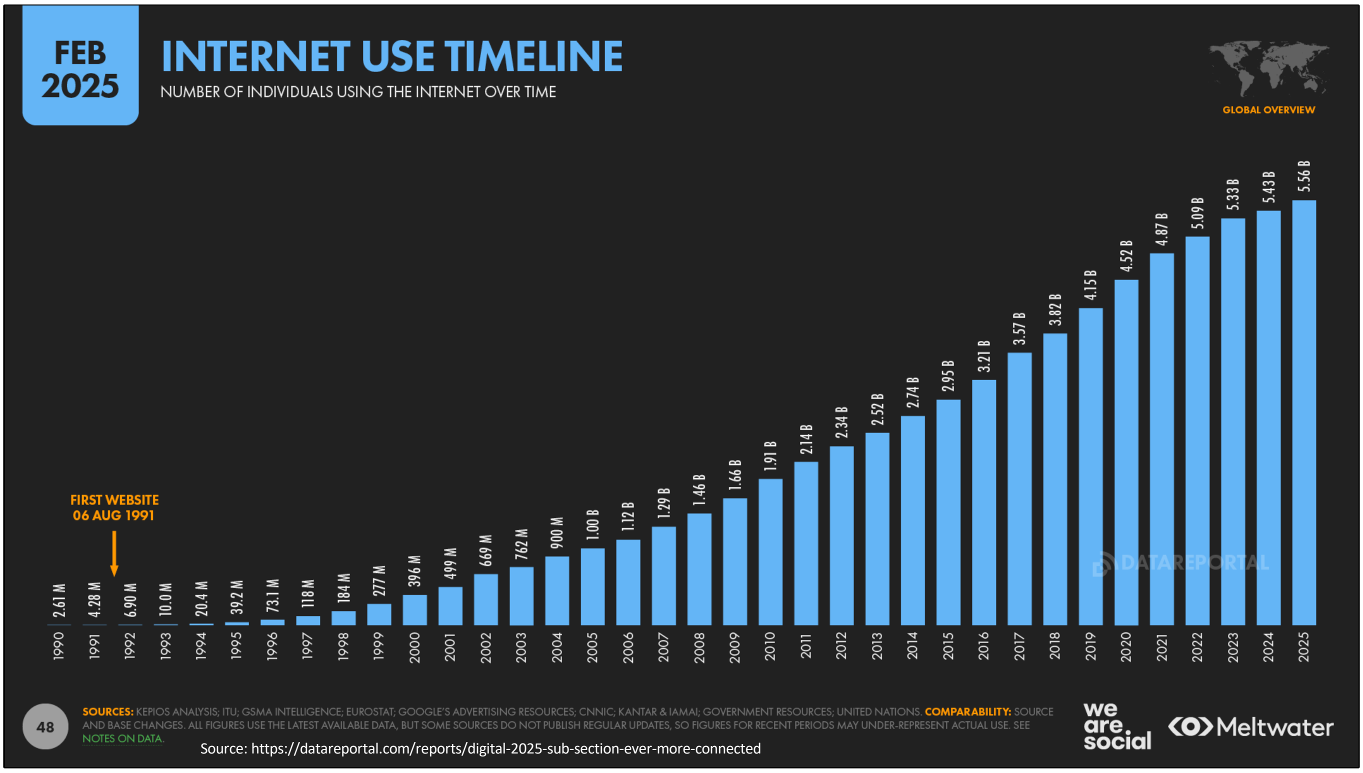

Source: https://datareportal.com/reports/digital-2025-sub-section-ever-more-connected

What do we use computer networks for?

Communication & Interaction

Entertainment & Media

Information & Education

Economy & Business

Services & Convenience

Course Practicalities

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Course organisation

- **Theory lectures**
  - 2 hours per week
  - Thursday 10h45 – 12h45 at M.A.143
  - Course material
    - Slides with notes (made available before each class via blackboard, without exercise solutions)
    - Optional material: Book (*Kurose and Ross, "Computer Networking: A Top-Down Approach", 8th Global Edition*)
  - Closed book exam with open questions (50% of total grade)
    - You can bring double-sided A4 paper with notes (written, printed, figures, text, etc.)
    - Example exam questions will be made available at the end of the semester
- **Lab sessions**
  - Computer-based exercises
  - Lab assignments made available after the associated theory chapter is completed
  - Contact moment every Friday 08:30 – 10:30 at M.G.025 + M.G.026
  - Possibility to work from home and/or on-campus
  - Solution handed in via Blackboard 2 weeks after assignment (50% of total grade)
  - In groups of two students

!! minimum requirement to pass!!

You must obtain 4/10 on both the theory exam and lab assignments

If you fail only 1 part, the grades of the other can be transferred to the re-examination during summer

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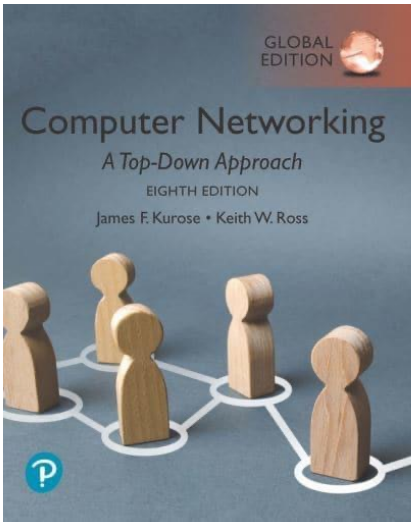
Optional course material

- **Kurose and Ross, "Computer Networking: A Top-Down Approach", 8th Global Edition**
  - Author website: https://gaia.cs.umass.edu/kurose_ross/index.php
  - Video lectures by James Kurose: https://gaia.cs.umass.edu/kurose_ross/lectures.php
  - Wireshark exercises: https://gaia.cs.umass.edu/kurose_ross/wireshark.php
  - Interactive exercises: https://gaia.cs.umass.edu/kurose_ross/interactive/

Course content

- Chapter 1: Computer Networks and the Internet
- Chapter 2: Application Layer
- Chapter 3: Transport Layer
- Chapter 4: Network Layer (Data Plane)
- Chapter 5: Network Layer (Control Plane)
- Chapter 6: Link Layer
- Chapter 7: Wireless and Mobile Networks
- Chapter 8: Network Security

Part of other courses in bachelor and master



Lab assignments

- Lab 0: Introduction to Mininet
- Lab 1: Application layer (HTTP)
- Lab 2: Transport layer (TCP/UDP)
- Lab 3: Network layer I (IPv4/IPv6)
- Lab 4: Network layer II (routing algorithms)
- Lab 5: Link layer I (LANs)
- Lab 6: Link layer I (MAC protocols) ➔ In person attendance required, solution handed in at the end of the session

Tentative course schedule (subject to change)

Week	Theory Lecture (Thursday @ 10h45)	Lab Session (Friday @ 8h30)
9 – 15 February	Part 0 (Practicalities) + Part 1 (Intro)	Lab 0: Introduction to Mininet
16 – 22 February	Part 1 (Intro) + Part 2 (Application Layer)	No lab
23 February – 1 March	Part 2 (Application Layer)	Lab 1 (Application Layer)
2 – 8 March	Part 3 (Transport Layer)	Lab 1 continued
9 – 15 March	Part 3 (Transport Layer)	Lab 2 (Transport Layer)
16 – 22 March	Part 4 (Network Layer)	Lab 2 continued
23 – 27 March	Part 4 (Network Layer)	Lab 3 (Network Layer I)
30 March – 5 April	Part 5 (Link Layer)	Lab 3 continued + Lab 4 (Network Layer II)
6 – 19 April	Holiday (Easter break)	Holiday (Easter break)
20 – 26 April	Part 5 (Link Layer)	Lab 4 continued
28 April – 3 May	Closing session & exam info	Holiday
4 – 10 May	Backup	Lab 5 (Link Layer I)
11 – 17 May	Holiday	Holiday
18 – 24 May	Lab 6 (Link Layer II)	Lab 6 continued

Homework: Preparation for the Labs

- During the lab sessions we will use a **Virtual Machine**.
- In order to prepare for the lab sessions, you need to download the **VM software** and the **VM itself** beforehand!

- For people using **Windows**, **Linux** or **MacOS on an Intel CPU:**
  - Download and install the **Virtualbox** software from https://www.virtualbox.org/
  - Also make sure to download and install the “Oracle VM VirtualBox Extension Pack
  - Download the actual VM image from https://student.idlab.uantwerpen.be/computernetwerken/Computernetwerken-2026-AMD64.ova
- For people using **MacOS on an Apple Silicon CPU:**
  - Download and install the **UTM** software from https://mac.getutm.app
  - Download the actual VM image from https://student.idlab.uantwerpen.be/computernetwerken/Computernetwerken-2026-ARM.utm.zip
- VM Username / Password: computernetwerken / mvkbj1n

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Learning outcomes

1. The student is able to determine to which layer a certain function or protocol of a computer network belongs.

2. The student has a view on the main functions, mechanisms, and protocols in the physical, data link, network, transport, and application layer of modern computer networks in general and the Internet in particular.

3. The student is able to implement computer programs that make use of these mechanisms and protocols to communicate across networks.

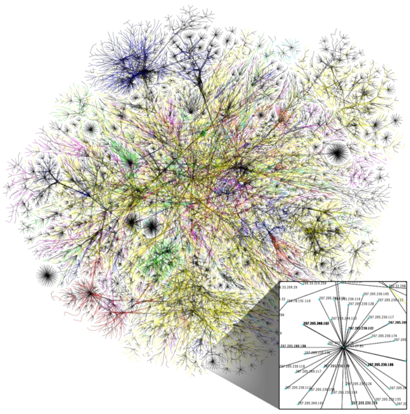
4. The student can properly configure computer networks, detect problems, and fix them.

Teaching approach

Top-down approach

- Application
- Transport
- Network
- Link
- Physical

Main focus on the Internet



Contact details

Course Lecturer

Prof. Jeroen Famaey
jeroen.famaey@uantwerpen.be

Teaching Assistants

Johan Bergs
johan.bergs@uantwerpen.be

Andrei Belogaev
andrei.belogaev@uantwerpen.be

In person meeting possible after email appointment in The Beacon (Sint-Pietersvliet 7, 2000 Antwerpen) or Campus Middelheim (Thursday for Jeroen or Friday for Johan and Andrei)

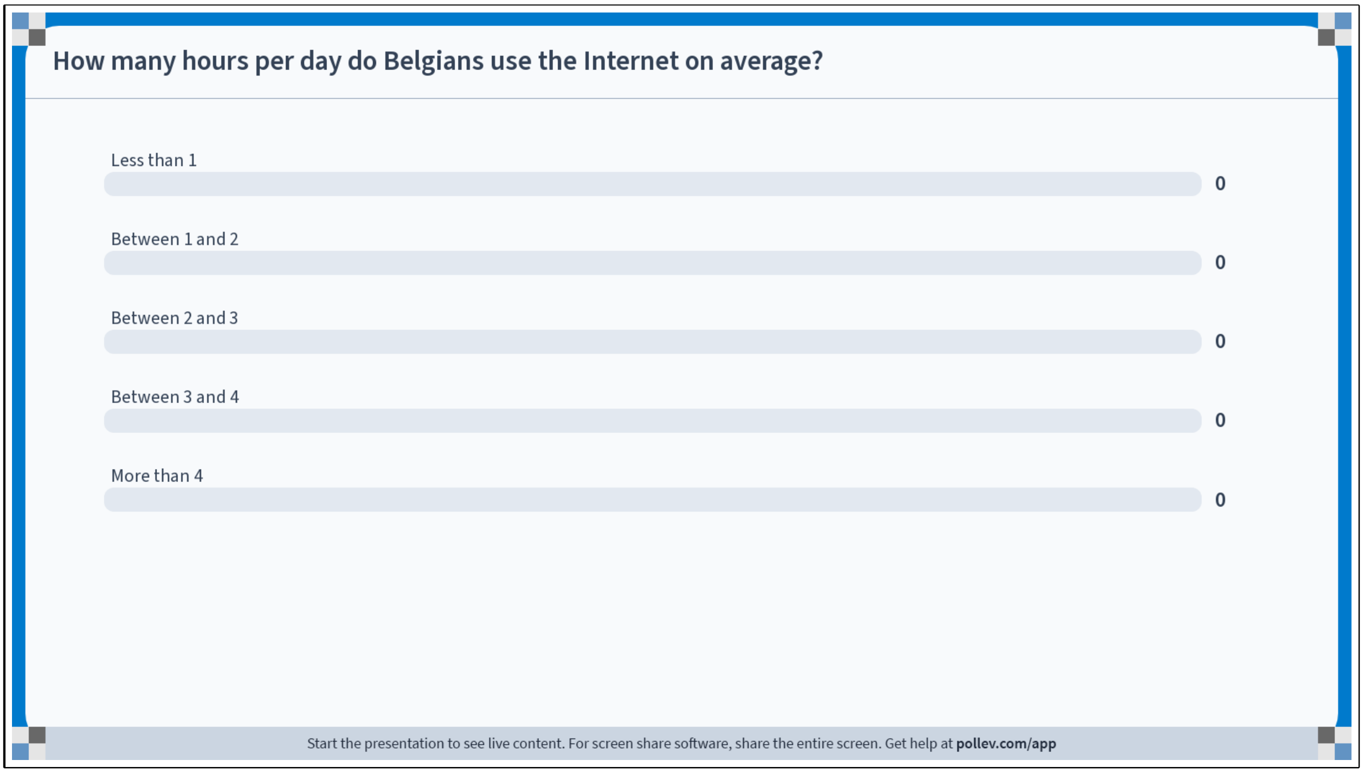
Related courses

Bachelor

- **Computer- en netwerkbeveiliging** (BA3, 1st semester) by Jeroen Famaey
- **Distributed Systems** (BA3, 2nd semester) by José Oramas

Master

- **Advanced wireless and 5G networks** (MA, 1st semester) by Jeroen Famaey
- **Future Internet** (MA, 1st semester) by Juan Felipe Botero
- **Introduction to performance modelling** (MA, 1st semester) by Benny Van Houdt
- **Advanced performance modelling** (MA, 2nd semester) by Benny Van Houdt
- **Advanced networking lab** (MA, 2nd semester) by Jeroen Famaey
- **Topics in computer networks** (MA, 2nd semester) by Miguel Camelo
- **Internet of Things** (MA, 2nd semester) by Miguel Camelo



Do not modify the notes in this section to avoid tampering with the Poll Everywhere activity.
More info at polleverywhere.com/support

How many hours per day do Belgians use the Internet on average?
https://www.polleverywhere.com/multiple_choice_polls/9EcRAaBXXbZ22ZjUmqva2?state=opened&flow=Default&onscreen=persist

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End of Part 0

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